Lab Study 3 (Group 1)

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draws = 500kW real power = 480V Pf init. of = 0,75 lagging $Z_{line} = 0{,}015 + j0{,}03\Omega$

Part 1)

$P_{load} = 500\ kW$ at $V_{LL} = 480V$ (assume 3-phase)

$P_f = 0{,}75$ lagging

feeder impedance per phase $Z_{line} = 0{,}015 + j\,0{,}033\Omega$ $R = 0{,}015\Omega$

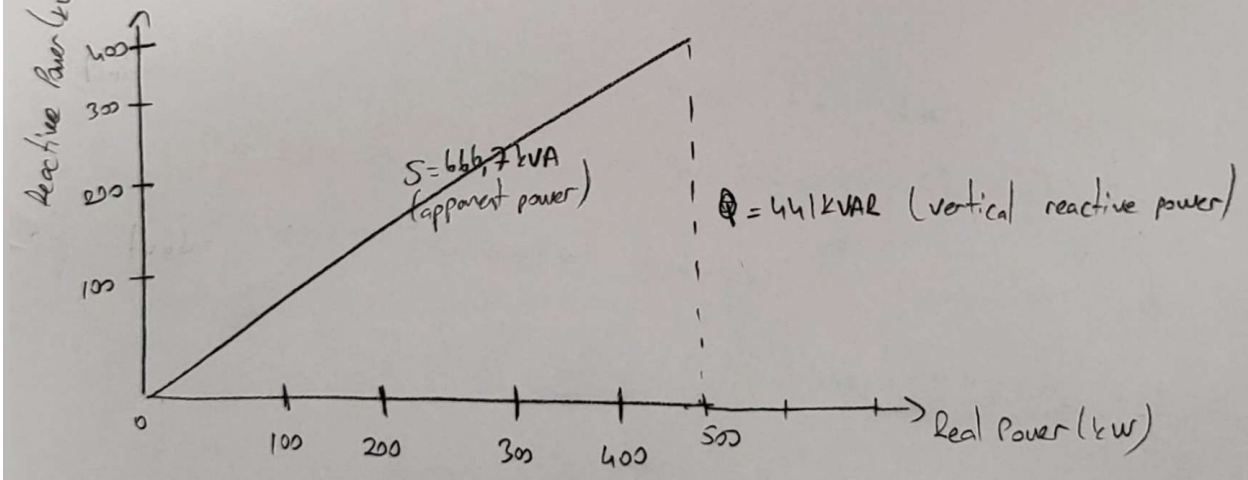
1) System loading $S_{initial}$ and $\Theta_{initial}$

$P \tan\phi$ $\phi = \cos^{-1}(pf)$

$S_{init.} = \frac{P}{pf} = \frac{500kW}{0{,}75} \approx 666{,}7\ kVA$

$\phi_{initial} = \cos^{-1}(0{,}75) \approx 41{,}41$

$\Theta_{initial} = P \tan\phi \approx 500 \times \tan(41{,}41°) \approx 441\ kVAR$



$P = 500kW$

$\phi = 441\ kVAR$

2) feeder line Current ($I_{line,initial}$)

$$I_{line,init.} = \frac{S_{initial}}{\sqrt{3}V_{LL}} = \frac{666.7 \times 10^3}{\sqrt{3} \times 480} \approx 802A$$

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3) Feeder $I^2 R$ Loss ($P_{loss,initial}$)

3 phase feeder is $3I^2 R$ per phase R

$P_{loss,initial} = 3 I_{line}^2 \cdot R = 3 \times (802A)^2 \times 0,015\Omega \approx 28,9 kW$

That's real power burned as heat in 480V feeder.

Part 2 - Cost - Benefit Justification (feasibility)

1a) With fixed kVA (or ampere) capability in transformers / cables, reactive current "uses up" capacity without producing real work. A low pf increases total current

$I = \frac{P}{\sqrt{3} V_{LL} pf}$ Pushing equipment closer to its thermal current limits.

This reduces the headroom for additional kW load even if voltage is available capacity spent carrying kVAR

1b)

a) what is feeder kVA pf=1 limit 700A?

$S_{max} = \sqrt{3} V_{LL} \cdot I_{max} = \sqrt{3} \times 480 \times 700 \approx 582 kVA \Rightarrow P_{max} (pf=1) \approx 582 kW$

b) How much extra real power compared 500kW?

$\Delta P_{max} \approx 582 - 500 \approx 82 kW$

2a) $P_{loss,initial}$

a) Energy charges (kWh) $\approx$ 28,9 kW

b) Demand/kVA & pf penalties? Higher current raises peak kVA and can trigger

2b) Real power kW is what plant needs to produce goods. P.factor correction targets kVAR, which cuts current, releases capacity, and slashes $I^2 R$ losses and demand charges without lowering production. Hence, the business case is about reducing kVAR not kW.

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